Domaine Dalachi: 10-year satellite zoning



Item	Value
Location	Zaghouan, Tunisia
Area	5.83 ha (measured from boundary coordinates)
Variety	Baldi (family variety)
Density	~100 trees/ha, rainfed (no irrigation)
Terrain	Mountain slope, 167–195 m altitude, descending from south-west (near a forest) to north-east (along a paved road); whole farm faces north; visible contour terraces

1. Data collected

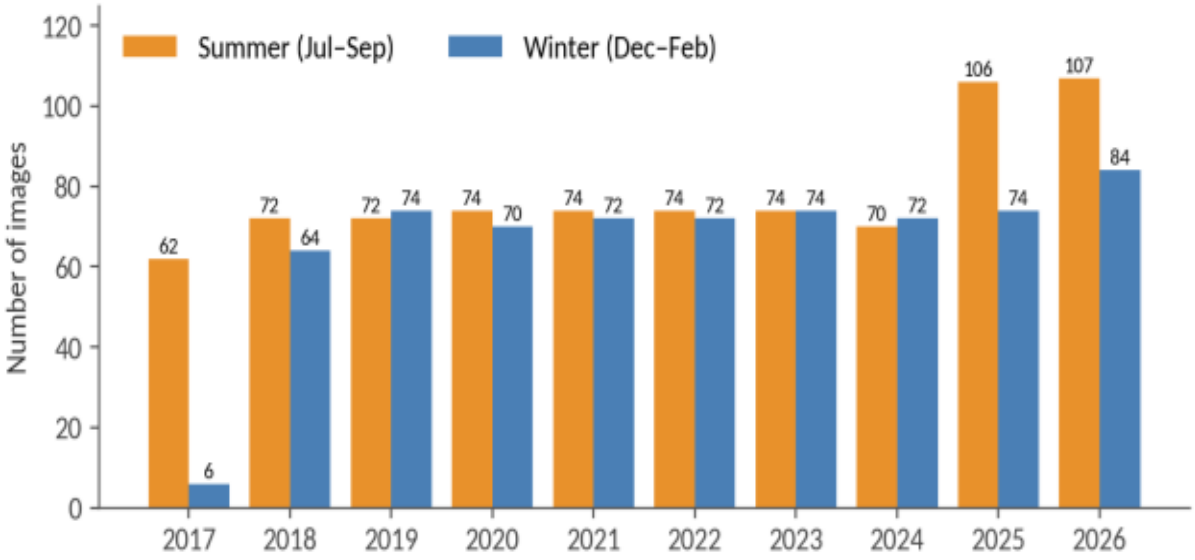


Figure: Number of Sentinel-2 images covering the farm in each season. Summer = July to September; winter = December of the previous year to February

After cloud masking, all 472 interior pixels had a complete ten-year history: no pixel had to be dropped for missing data. The 2026 summer composite covers July to late September, a few days short of the full season; because a median is used, this has a negligible effect.

The farm's interior (4.73 ha, once the 10 m edge strip is removed) was cut into 472 pixels of 10 m × 10 m. Each pixel was then described by five numbers, called features, that summarise how it behaved over the ten years (Table 4). Four are based on summer, when the trees dominate the green signal; one looks at the 2023 drought, the driest recent year.

Feature	How it is calculated	What it describes	Farm average
Summer NDVI	Median of the ten summer NDVI values	Typical summer vigour (green canopy)	0.173
Summer NDMI	Median of the ten summer NDMI values	Typical summer canopy water signal (strongly influenced by soil)	−0.090
Summer NDRE	Median of the ten summer NDRE values	Typical summer chlorophyll	0.119
Seasonal swing	Winter NDMI minus summer NDMI, median over the years	How much the patch 'breathes' between the wet and the dry season	0.089
2023 drought response	2023 summer NDMI minus the median of the other summers	How much drier than usual the patch was in the 2023 drought (more negative = hit harder)	−0.023

Table 4. The five features describing each pixel, with their average over the farm.

Statistic	NDVI	NDMI	NDRE	Swing	2023
Mean (average)	0.173	−0.090	0.119	0.089	−0.023
Standard deviation	0.026	0.014	0.018	0.029	0.010
Minimum	0.102	−0.120	0.073	0.052	−0.048
25th percentile	0.158	−0.100	0.106	0.072	−0.029
Median (50th)	0.174	−0.092	0.120	0.085	−0.023
75th percentile	0.189	−0.082	0.132	0.096	−0.016
Maximum	0.281	−0.028	0.187	**0.294**	−0.001

Table 5. Distribution of the five features across the 472 pixels.

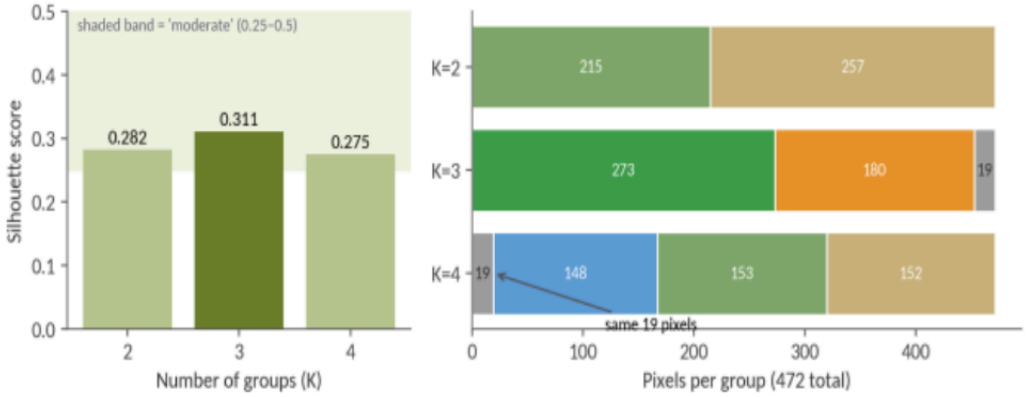
Most of the farm is fairly similar, but one number stands out: the maximum seasonal swing (0.294) is more than three times the average (0.089). A handful of pixels green up and dry out far more strongly than the rest. They turn out to be the small Z3 patch described in Section 6.3.

2. Finding the zones

- How pixels were grouped: k-mean clustering

K-means works in four simple moves. First, each feature is standardized: rescaled so that a difference of one standard deviation counts the same for every feature; otherwise, a feature measured in larger numbers would dominate. Second, each pixel is imagined as a point in a five-dimensional space, one dimension per feature; pixels that behave alike sit close together. Third, K 'centres' are placed and every pixel is assigned to its nearest centre. Fourth, each centre moves to the middle of its pixels, and the last two moves are repeated until nothing changes. The procedure was run 20 times from different starting points and the best result kept.

A measure of how well separated the groups are. For each pixel it compares how close the pixel is to its own group with how close it is to the nearest other group, then averages over all pixels. It ranges from −1 to +1: above 0.5 = clearly distinct groups; 0.25–0.5 = real but moderate differences; below 0.25 = weak structure.



Groups (K)	Silhouette	Pixels per group	Reading
2	0.282	215 / 257	The 19 unusual pixels are forced into a main group, which becomes a poorer fit
3	0.311	273 / 180 / 19	Best: the unusual pixels get their own group and the rest splits cleanly in two
4	0.275	19 / 148 / 153 / 152	Same 19 pixels; the main area is split one extra time with no gain in clarity

All three scores are moderate (0.28–0.31). This means the farm changes gradually from one part to another rather than having sharp borders; the zones are real, but their edges are soft. A moderate score is common on natural terrain and is not a sign of failure, provided the zones prove stable over time (Section 8).

K-means pays no attention to the size of a group, only to how similar its members are. When a small set of pixels behaves very differently from everything else, the method gives them their own group, because forcing them into a larger group would give that group a poor description of its members. Here, 19 pixels have a seasonal swing roughly double that of the rest of the farm. Their reappearance as a separate group at both K = 3 and K = 4 shows that they are a genuine, distinct feature of the data, not a random artefact.

- Cleaning the map

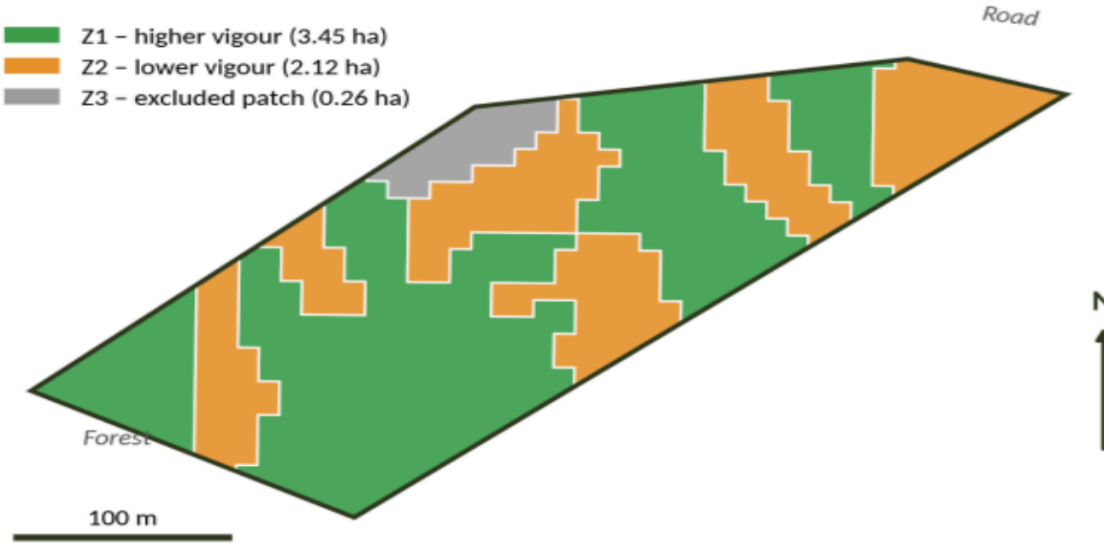
The raw result of k-means is speckled, because the method ignores neighbours. Three cleaning steps turned it into a usable map:

A) Majority filter. Each pixel looks at its eight neighbours and switches to the group that clearly dominates around it. Isolated speckles disappear.

B) Fragment merge: Any patch smaller than 8 pixels (0.08 ha) joins the neighbouring zone it touches most; a patch of three or four trees cannot be managed separately.

C) Edge fill. Pixels in the 10 m edge strip, left out of the grouping, take the zone of their nearest neighbour so that the zones cover the entire farm.

Cleaning moved a few pixels between groups: the raw counts of 273 / 180 / 19 became 284 / 171 / 17. Thezones were then numbered by their summer NDMI, from lowest (Z1) to highest (Z3).



Zone	Area	Summer NDVI	Summer NDMI	Summer NDRE	Seasonal swing	2023 response
Z1	3.45 ha	0.185	−0.096	0.128	0.092	−0.025
Z2	2.12 ha	0.150	−0.082	0.101	0.074	−0.019
Z3	0.26 ha	0.213	−0.063	0.138	**0.189**	−0.026

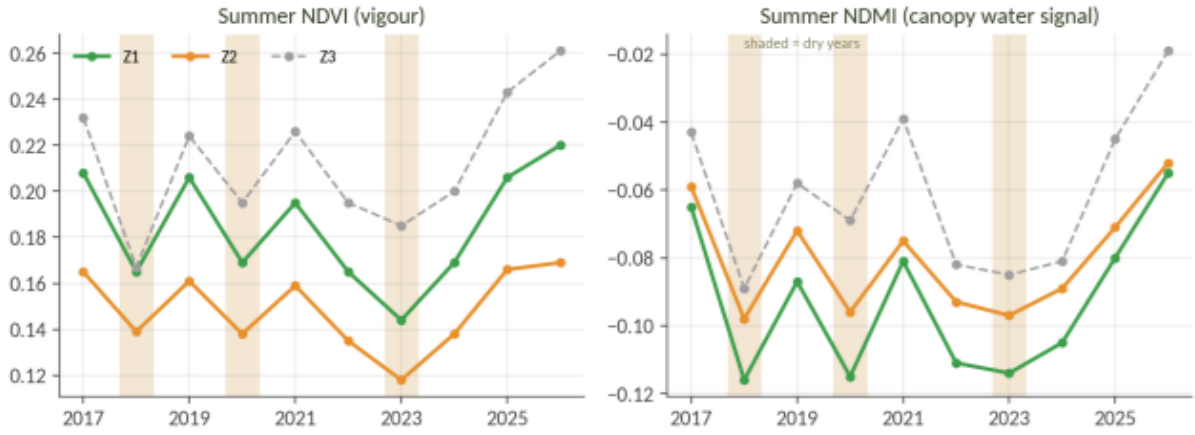
Z1: carries more green canopy (NDVI 0.185) and more chlorophyll (NDRE 0.128) than Z2. It is made of four separate pieces, the largest covering the south-west and centre of the farm. Its summer NDMI is the lowest of the three zones, which at first sight suggests a 'drier' canopy. Because each pixel is mostly soil (Section 2.4), this is more likely an effect of the ground than of tree stress. The most reliable description of Z1 is therefore 'more green canopy', not 'drier'.

Z2: shows less green canopy (NDVI 0.150) and less chlorophyll (NDRE 0.101), and the smallest seasonal swing. It forms six strips and blocks that, on the satellite image, run across the contour terraces, down the slope. Section 9 examines whether these strips correspond to small ridges that shed rainwater.

Z3 is a single patch of 17 pixels in the north-west corner, touching the road at one end. Its seasonal swing (0.189) is twice that of the rest of the farm: it greens up strongly in winter and dries out hard in summer. This is the typical behaviour of annual vegetation (grass, weeds) or of a spot where water collects. On the high-resolution image the ground there looks lighter and a few larger trees are visible. Whatever the cause, it does not behave like the rest of the grove, so it is excluded from farm averages and from the choice of sampling trees, while being kept under observation.

- Are the zones stables?

Year	NDVI Z1	NDVI Z2	NDVI Z3	NDMI Z1	NDMI Z2	NDMI Z3	Same ranking?
2017	0.208	0.165	0.232	−0.065	−0.059	−0.043	Yes
2018	0.165	0.139	0.167	−0.116	−0.098	−0.089	Yes
2019	0.206	0.161	0.224	−0.087	−0.072	−0.058	Yes
2020	0.169	0.138	0.195	−0.115	−0.096	−0.069	Yes
2021	0.195	0.159	0.226	−0.081	−0.075	−0.039	Yes
2022	0.165	0.135	0.195	−0.111	−0.093	−0.082	Yes
2023	0.144	0.118	0.185	−0.114	−0.097	−0.085	Yes
2024	0.169	0.138	0.200	−0.105	−0.089	−0.081	Yes
2025	0.206	0.166	0.243	−0.080	−0.071	−0.045	Yes
2026	**0.220**	**0.169**	**0.261**	**−0.055**	**−0.052**	**−0.019**	Yes



The ranking never changed. Every year, for both indices, the zones kept the same order. This is the strongest result of the whole analysis: the zones are permanent features of the farm. All zones move together with the weather. Dry years (2018, 2020, 2022–2024) pull the whole farm down; better years (2021, 2025, 2026) lift it. Weather acts on the farm as a whole. The gaps between zones stay roughly constant. Whatever the year, Z1 stays about 0.03–0.05 NDVI

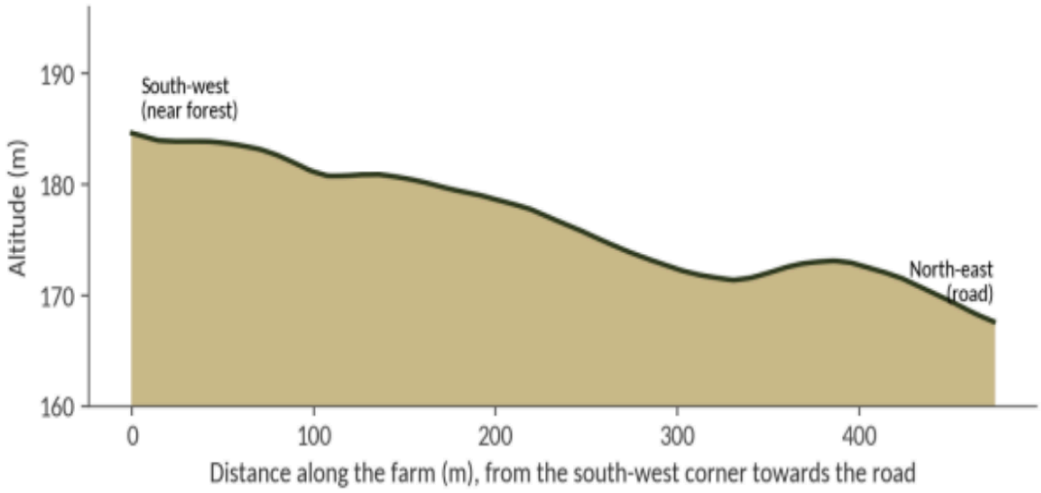
above Z2. Year-to-year swings are larger than the differences between zones. This matters for monitoring: to detect a problem, each zone should be compared with its own history rather than with the other zones.

2026 is the greenest year of the record for every zone, continuing a rise that started in 2025. This may reflect a favourable water year and should be checked against weather data, since it may influence this season's harvest.

- What the terrain tells us

Domaine Dalachi lies on a mountain slope, so its trees are not all at the same altitude and rainwater does not stay where it falls: it runs downhill, collects in hollows and drains away from bumps. On a rainfed farm, this redistribution of water can be the main reason some trees grow better than others. The zones were therefore compared with the shape of the land.

The farm descends from about 193 m near the forest in the south-west to about 167 m along the road in the north-east: a drop of roughly 30 m over 500 m (Figure 9). The slope is steepest in the south-west (up to 8–9°) and becomes gentler towards the road. The whole farm faces north (northness 0.8–1.0), which is favourable for rainfed olives: less direct sun and slower drying of the soil.



3. Tree Selection

We selected the trees based on the same rules:

- inside the right zone, and at least 15–20 m from any zone border
- at least 30 m from the road and from Z3
- at least 20 m from the farm edge, and at least 40 m from the forest corner
- then the best terrain for each role

Recommended sampling locations (dashed circle = 15 m search area)



Tree	Role	Coordinates	Zone	Altitude	Landform	Rainwater received (farm median = 6)
TREE_01	**Receiving**		Z1	172 m	Clear hollow (TPI −1.06)	**200**
TREE_02	**Intermediate**		Z1	185 m	Neutral (TPI +0.02)	6 (typical)
TREE_03	**Drainant**		Z2	171 m	Clearest bump (TPI +0.54)	**3** (sheds water)